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Exploring the Impact of Generative AI ChatGPT on Critical Thinking in Higher Education: Passive AI-Directed Use or Human–AI Supported Collaboration?

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Abstract

Generative AI is weaving into the fabric of many human aspects through its transformative power to mimic human-generated content. It is not a mere technology; it functions as a generative virtual assistant, raising concerns about its impact on cognition and critical thinking. This mixed-methods study investigates how GenAI ChatGPT affects critical thinking across cognitive presence (CP) phases. Forty students from a four-year university in the southwestern United States completed a survey; six provided their ChatGPT scripts, and two engaged in semi-structured interviews. Students' self-reported survey responses suggested that GenAI ChatGPT improved triggering events (M = 3.60), exploration (M = 3.70), and integration (M = 3.60); however, responses remained neutral during the resolution stage. Two modes of interaction were revealed in the analysis of students' ChatGPT scripts: passive, AI-directed use and collaborative, AI-supported interaction. A resolution gap was identified; nonetheless, the interview results revealed that when GenAI ChatGPT was utilized with guidance, all four stages of cognitive presence were completed, leading to enhanced critical thinking and a reconceptualization of ChatGPT as a more knowledgeable other. This research suggests that the effective use of GenAI in education depends on the quality of human–AI interaction. Future directions must orient toward an integration of GenAI in education that positions human and machine intelligence not as a substitution but as co-participation, opening new epistemic horizons while reconfiguring assessment practices to ensure that human oversight, critical inquiry, and reflective thinking remain at the center of learning.

Keywords: generative AI; ChatGPT; critical thinking (CT); practical inquiry (PI); cognitive presence (CP); community of inquiry (CoI); GenAI-directed; GenAI-supported; higher education

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1. Introduction

Every generation witnesses anxiety about new educational tools. Spelling and grammar checkers and calculators were once feared as threats to learners' cognitive development;

over time, however, they became normalized as invaluable aids. Today, generative AI (GenAI), most visibly ChatGPT, introduced in 2022, has prompted similar discussions.

ChatGPT features significant potential in education by facilitating personalized learning experiences and enabling interactive question-and-answer conversations across diverse subjects (AlAli & Wardat, 2024), providing real-time feedback (Birenbaum, 2023), and functioning as effective tutors that assist students in solving complex problems and fostering independent learning (Jin et al., 2024). However, educators remain apprehensive that the accessibility of AI-generated responses may result in metacognitive laziness and poor cognitive engagement, consequences that in turn may negatively affect reasoning (Costa et al., 2024), negatively impact problem-solving abilities (Rahman & Watanobe, 2023), and ultimately promote academic dishonesty (Neumann et al., 2023).

Yet, despite these opportunities and risks, a substantive research gap persists: There is limited understanding of whether ChatGPT fosters critical inquiry or engenders cognitive offloading. Addressing this gap is critical in higher education, where critical thinking is regarded not merely as a core learning outcome but also as a fundamental basis for meaningful engagement with complex ideas. An active learner capable of analyzing, evaluating, and synthesizing information is central to the learning process (Greenhill, 2010), and this emphasis is echoed in modern educational standards, such as the 21st century's C4 framework of critical thinking, creativity, communication, and collaboration.

Thus, this study adopts the practical inquiry cognitive presence (PICP) model as a theoretical lens to examine how ChatGPT affects critical thinking (Breivik, 2016; Kaczkó & Ostendorf, 2023; Olivier & Weilbach, 2024; Yang & Mohd, 2020). Cognitive presence captures the extent to which learners construct meaning through reflection and discourse, aligning closely with critical thinking inquiry. The model includes four stages: triggering event, exploration, integration, and resolution (Garrison et al., 2001). The fundamental critical thinking activities (questioning assumptions, gathering information, synthesizing concepts, and reaching a conclusion/applying solutions) are mapped onto each phase, which correspond to distinct thinking behaviors ranging from divergent idea generation to convergent problem-solving. For example, a triggering event presents a puzzling issue that prompts questions and curiosity; exploration involves brainstorming and information gathering; integration entails organizing and constructing meaning from new insights; and resolution involves breaking down complexities and applying the knowledge in practice.

The PICP model therefore provides a structured lens for analyzing students' interactions and engagement with ChatGPT and for organizing how students can utilize ChatGPT in collaborative, AI-supported learning. This mixed methods study investigates students' perspectives on the influence of GenAI ChatGPT on critical thinking in light of the PI's cognitive presence phases. Are their ChatGPT interactions passive, such as accepting AI responses on the premise that they are correct by nature, or active, like synthesizing concepts? What if students follow the steps of cognitive presence as a critical thinking pathway to guide their interaction with GenAI ChatGPT? We argue that a human-in-the-loop approach, where learners remain actively reflective and evaluative in their GenAI interactions by applying their critical thinking, is essential. Using the PICP model, this study operationalizes critical thinking as a measurable process to examine whether GenAI fosters or hinders its stages.

2. Literature Review

2.1. What Higher Education Gains and Risks in the Age of Generative AI

GenAI tools have rapidly integrated into classrooms and online learning environments. Since its release in November 2022, students and educators alike have engaged in an ongoing experiment with ChatGPT's possibilities, testing how such a tool might extend or

constrain the learning process. The evolution of ChatGPT to produce human-like content and dialogues distinguishes it from all other technical tools. It symbolizes more than a technology advance; it marks a turning point that challenges educators and institutions to rethink habits, reshape methods, and develop new teaching and learning strategies. Thus, UNESCO and the European Parliament have promoted GenAI integration into education and ethical AI use by students and instructors (Cukurova & Miao, 2024; European Parliament, 2024). In fact, GenAI provides significant advantages for both educators and students by saving the time teachers spend on administrative duties (Cukurova & Miao, 2024; European Parliament, 2024). It provides immediate feedback, thereby decreasing cognitive load and allowing students to focus on essential and innovative aspects of learning (Chang et al., 2024; Dai et al., 2023). This time-saving enables educators and students to concentrate on enhancing students' creative and analytical skills while effectively managing classroom dynamics. Moreover, GenAI assists disadvantaged and special needs students by enhancing academic achievement, promoting inclusivity, and providing access to educational experiences tailored to their specific needs.

However, the integration of GenAI into educational environments raises substantial challenges. Its application raises ethical concerns, especially with algorithmic bias, as training data may reinforce existing social preconceptions. As students increasingly rely on AI-generated content, apprehensions arise regarding the deterioration of critical thinking and academic integrity. Overdependence on GenAI like ChatGPT may foster plagiarism and undermine learners' capacities for original thought and creative expression (Essien et al., 2024; Santiago et al., 2023). Moreover, GenAI may pose an additional hazard that might infringe upon users' privacy, facilitate data colonization, and be weaponized against the user. In summary, GenAI in higher education provides significant new capabilities, immediate flexibility, individualized instruction, and enhanced creative opportunities, but also revises enduring discussions around reliance on technology. The literature recognizes the pros and cons of GenAI use. On the upside, recent studies report that using GenAI within Educational Escape Rooms (EERs) has demonstrated a reduction in cognitive load and an improvement in critical thinking (Fotaris et al., 2023). Also, GenAI chatbots have shown increased cognitive presence within the community of inquiry framework by supporting analytical engagement (Olivier & Weilbach, 2024). On the other hand, GenAI's tendency to generate inaccurate information suggests the possibility that its uncritical adoption could lead to shallow learning outcomes (Rahman & Watanobe, 2023). Surveys conducted among educators and learners reveal differing perspectives: Many participants view AI as a valuable educational tool, whereas others express apprehensions regarding its long-term effects on the quality of education (Valova et al., 2024). Lodge et al. (2023) and Chan and Hu (2023) emphasize the importance of combining human collaboration with AI to provide a comprehensive educational experience and call for studying its effects on learning processes, especially critical thinking (see Figure 1).

2.2. GenAI Impact on Critical Thinking

Critical thinking is a multidisciplinary core skill and one of the C4 competencies for the 21st century that might represent a potential solution to current challenges in both education and society. Critical thinking in higher education allows students to analyze material critically and engage in abstract reasoning, transcending the mere acquisition of isolated knowledge (Greenhill, 2010). The process includes investigating, evaluating, and synthesizing information, leading to well-founded decisions (Akbar, 2023; Baird & Parayitam, 2019). In addition, educational standards such as the ISTE standards for students emphasize the role technology plays in strengthening these skills while promoting ethical

behavior and responsible digital citizenship in AI-driven learning environments (ISTE 1.1.c; Digital Citizen 2.b; Crompton & Burke, 2024).

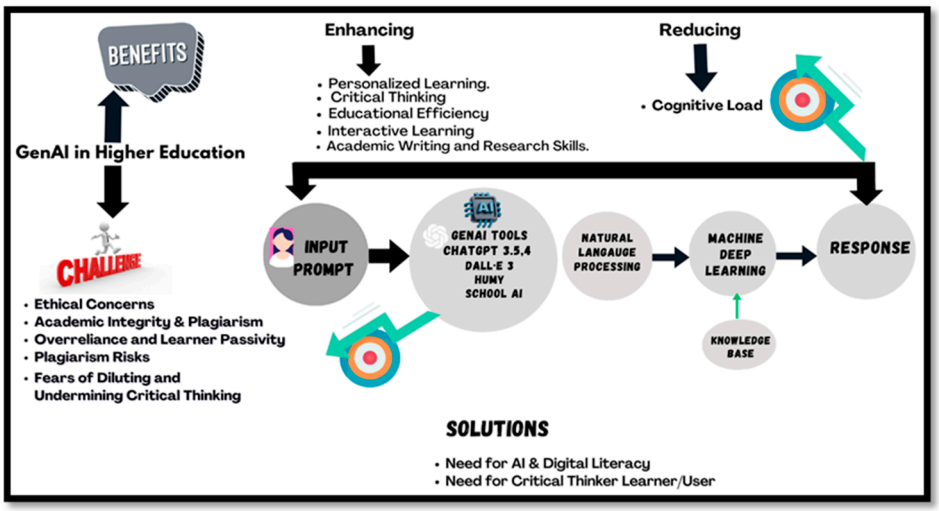

Figure 1. Overview of GenAI in higher education: benefits, challenges, and solutions. Note: This figure emphasizes the dual contributions of GenAI in higher education, derived from literature review analysis, highlighting initiatives that utilize its advantages while addressing its related threats (Nasr et al., 2025).

Therefore, the investigation of critical thinking and AI literacy among learners and educators addresses the necessity for the human-in-the-loop method and highlights the significance of maintaining human oversight in the implementation of AI within educational contexts. Critical thinking and AI literacy are essential competencies for the responsible and effective utilization of GenAI in educational contexts (European Parliament, 2024). However, the impact of GenAI on critical thinking is intricate. Despite the positive effect of GenAI on critical thinking found in several studies, there are ongoing concerns regarding the possible overreliance on AI-generated knowledge, which might undermine critical and creative thinking.

The influence of GenAI on critical thinking can pose as both benefits and obstacles in the educational environment. GenAI technology enhances critical thinking by improving content engagement (Alam, 2022). Also, ChatGPT's functions facilitate students' critical assessment of AI-generated materials in terms of credibility and dependability. The development of the ChatGPT Literacy Scale by Lee and Park (2024) further supports the notion that AI can be instrumental in teaching students to assess digital content with a critical lens.

Furthermore, research across various disciplines, including nursing and STEM education, indicates that the integration of ChatGPT leads to notable enhancements in critical thinking, problem-solving, and learning outcomes (Chang et al., 2024; Li et al., 2024). Educational Escape Rooms (EERs) seem to reduce cognitive load and enhance critical thinking (Fotaris et al., 2023). The use of GenAI chatbots improves cognitive presence within the community of inquiry framework by promoting critical thinking (Olivier & Weilbach, 2024).

Some studies, however, indicate that GenAI may hinder students' creative and critical thinking by offering prefabricated answers, thereby reducing cognitive engagement and leading to cognitive offloading (Nasr et al., 2025). Kaczkó and Ostendorf (2023) discuss the challenges associated with promoting genuine critical thinking in AI-driven educational environments and suggest that the integration of AI models within the community of inquiry framework may present operational difficulties. In light of the concerns regarding the reliability and completeness of AI-generated responses, especially in high-stakes examinations, Freire et al. (2023) indicate that GenAI ought to be utilized as a supplemen-

tary tool rather than as a primary source of knowledge. Mixed perceptions regarding the efficacy of ChatGPT highlight the necessity for more balanced and empirical studies to comprehensively assess its long-term impact on critical thinking in education (Valova et al., 2024). Also, Lodge et al. (2023), Chan and Hu (2023), and Nasr et al. (2025) emphasize the importance of combining human collaboration with AI for a comprehensive educational experience and advocate for longitudinal studies on the ethical and pedagogical ramifications of GenAI utilization.

3. Theoretical Framework

We adopt Garrison et al.'s (2001) practical inquiry (PI)'s cognitive presence to examine critical thinking. Cognitive presence is defined as "the extent to which learners are able to construct and confirm meaning through sustained reflection and discourse" (Garrison et al., 2001, p. 11). This definition explicitly ties cognitive presence to critical inquiry: Knowledge is built via reflective problem-solving discussions. Garrison and Archer (2000) further defined cognitive presence from the perspective of the Practical Inquiry (PI) model and integrate it with traditional critical thinking processes conceptualized in Dewey's reflective thought (Dewey, 1933; Garrison & Akyol, 2013) and Lipman's community of inquiry (Lipman, 1997).

The PI framework organizes critical inquiry into four interrelated phases. A triggering event arises and leads to a complex issue or quandary. Then it creates cognitive dissonance, which prompts learners to recognize a deficiency in understanding (Garrison et al., 1999; Garrison & Vaughan, 2008). In a GenAI context, this may be presented as a sequence in which a student poses a query to ChatGPT or receives an unexpected response; then, their curiosity is stimulated.

The next step is exploration marked by multiple approaches to knowledge acquisition and idea generation (Garrison et al., 1999; Rolim et al., 2019). Students autonomously or cooperatively investigate resources, including refining prompts for the AI, to obtain ideas, theories, or viewpoints Here, learners "float" between private reflection and group (or AI-facilitated) discourse, testing out thoughts without judgment.

In the third phase, integration is a convergent process in which students systematically organize and synthesize their exploratory results into a coherent understanding (Garrison & Vaughan, 2008; Garrison et al., 2001). They rigorously assess evidence, such as verifying GenAI's responses against their prior knowledge, social schema, or external sources, and begin to formulate credible answers. This demonstrates the use of advanced cognitive processes, such as connecting ethical issues, avoiding biases, and correcting misinformation.

Finally, in the resolution phase, learners break down complexity and apply their newly learned knowledge in practical contexts, such as hands-on exercises, debates, or projects, to assess its validity (Garrison, 2017; Garrison & Vaughan, 2008). A successful resolution reinforces the inquiry cycle by addressing the triggering issue or creating new questions, then results in a deeper level of investigation. Appling this to AI, this involves contextualizing generated outputs, verifying their accuracy, and using them meaningfully in authentic tasks such as projects, debates, or problem-solving.

The PI model is undergirded by two conceptual axes: perception–conception and action–deliberation (Garrison, 2017). The perception–conception axis spans the continuum from initial perception (i.e., intuition and divergent thinking) to refined conception (i.e., convergence and reasoning). Creative leaps and brainstorming happen on the perception end, while logical analysis occurs toward conception. The action–deliberation axis encompasses reflective private thought and interactive social discourse. Learning progresses through the quadrants established by these axes.

The suitability of the Practical Inquiry model for this research lies in its extensive validation as a structured framework for evaluating critical thinking in online learning environments (Garrison et al., 2001; Breivik, 2016). This enables educators to monitor student progression through inquiry phases, including integration and resolution, thereby offering insights to students regarding their cognitive processes. Recent studies have investigated the PI model with GenAI and suggest that tools such as ChatGPT can improve cognitive presence by facilitating reflection and providing feedback in situations with scarce human guidance (Olivier & Weilbach, 2024).

With this updated model (see Figure 2), we align our analysis with emerging findings and fill a gap: While previous research has shown that GenAI promotes exploration and idea synthesis (Gregorcic et al., 2024; Olivier & Weilbach, 2024), we look specifically at how students engage with each phase in ChatGPT-supported tasks. Our focus on PICP phases thus provides a fine-grained lens on critical thinking: It allows us to see whether ChatGPT use leads learners to articulate new questions (triggering), probe ideas in dialogue with the AI (exploration), make connections and reason (integration), and reach a conclusion/apply solutions (resolution).

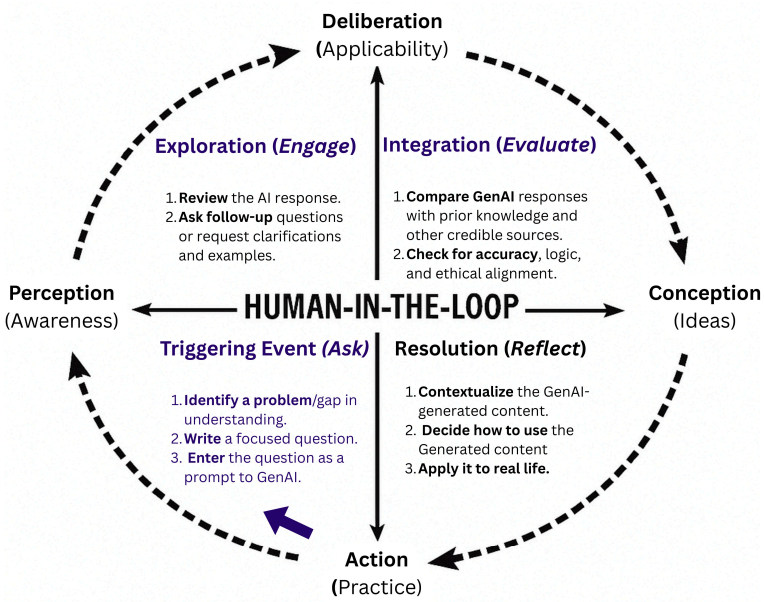

Figure 2. Adapting the Practical Inquiry framework to support critical thinking in a GenAI context. Note: Adapted from the Practical Inquiry Model by Garrison et al. (1999), Critical inquiry in a text-based environment: Computer conferencing in higher education, The Internet and Higher Education, 2(2–3), 87–105. This adaptation integrates cognitive presence staged to guide GenAI use to support critical thinking.

In sum, cognitive presence (in the PI model) offers a structured way to define and measure critical thinking in online GenAI-mediated learning. By tracing students' prompts, reflections, and chat responses through the four phases and two axes, we can characterize their role as a human in-the-loop. Accordingly, we can shed light on whether students are merely passively reading AI output and adopting it, or whether they are generating follow-up questions, critiquing the AI, and iteratively refining their understanding until they contextualize and practically refine the generated outcome to be applied in their learning situations.

Research Questions

1. What are students' self-reported perceptions of how Generative AI enhances their critical thinking?
2. To what extent do students demonstrate critical thinking while using Generative AI?

3. How do students perceive the impact of guided GenAI ChatGPT on their critical thinking?

4. Methodology

4.1. Design

The present study employed an explanatory mixed-methods design within a pragmatic framework to investigate higher education students' perceptions of GenAI ChatGPT and its impact on critical thinking.

4.2. Participants

Participants were higher education students enrolled in a four-year college of education at a southwestern U.S. university. Eligibility required that students be 18 years or older, have used ChatGPT at least once, and voluntarily agree to participate. Recruitment was conducted through departmental announcements and course mailing lists, resulting in 40 completed surveys. Because invitations were distributed broadly across departmental channels, the precise response rate could not be calculated.

Data collection included a 5-point Likert scale survey completed by all 40 participants, ChatGPT scripts from 6 students, and 45 minutes interviews with 2 of those students. Participant demographics are reported in Tables 1–3. While the qualitative data are not generalizable, they were intended to complement and enrich the survey findings.

Table 1. Participants' quantitative demographics.

Demographic	Number	Percentage
Total participants	40	100%
Gender (female)	35	87.50%
Gender (male)	5	12.50%
Age (18–24)	25	62.50%
Age (25–34)	8	20%
Age (35–44)	4	10%
Age (45–54)	3	7.50%
Ethnicity	White: 27, Asian: 5, Hispanic/Latino: 3, Other: 2, African American: 1, Native American/Alaska Native: 2	67.5% White, 12.5% Asian, 7.5% Hispanic/Latino, 5% Other, 2.5% African American, 5% Native American/Alaska Native
Academic level	Undergraduate: 25, graduate: 15	62.5% Undergraduate, 37.5% graduate
Disciplines	Elementary Education: 21, Early Childhood Education: 5, Curriculum and Instruction: 9, ESL/Bilingual Education: 3, Educational Technology: 1, Linguistics: 1	Elementary Education (52.5%; 21 students), Early Childhood Education (15%; 5), Curriculum and Instruction (22.5%; 9), ESL/Bilingual Education (7.5%; 3), Educational Technology (2.5%; 1), and Linguistics (2.5%; 1).

Table 2. ChatGPT script participants' demographics.

Demographic	Number	Percentage
Total participants	6	100%
Gender (female)	6	100%
Educational background	Undergraduate/Elementary education	100% Undergraduate
Age (18–24)	6	100%
Ethnicity (White)	6	100%

Table 3. Interview participants' demographics.

Demographic	Number	Percentage
Total participants	2	100%
Gender (female)	2	100%
Educational background	Undergraduate/Elementary education	100%
Age (18–24)	2	100%
Ethnicity (White)	2	100%

4.3. Data Collection

The study incorporated quantitative data from a 5-point Likert scale survey and employed a qualitative design using two complementary data sources: students' ChatGPT interaction scripts and semi-structured interviews. To protect confidentiality, all participants were assigned pseudonyms.

4.3.1. GenAI Critical Thinking Survey

A five-point Likert scale critical thinking survey was developed based on the Practical Inquiry Cognitive Presence Survey (Garrison & Arbaugh, 2007) and administered to a group of 40 participants. The purpose of the survey was to examine participants' perceptions of using GenAI ChatGPT for learning tasks and its perceived influence on critical thinking. The survey (See Supplementary Materials) was hosted on Google Forms for five weeks, from November to January 2023. Participation was voluntary. It consisted of four sections; each aligned with a key phase of the cognitive presence framework in the context of participants' interactions with ChatGPT.

4.3.2. ChatGPT Scripts

Without any instructional guidance, six elementary education undergraduates used ChatGPT to complete real coursework tasks (preparing a lesson plan). Each participant's chat history was organized and coded into four dimensions: triggering events, investigation, integration, and resolution. The coding system followed a strict framework, with no additional codes created or merged (Saldaña, 2021). The first author coded the transcripts, the third author reviewed them, and inter-rater reliability using Cohen's Kappa in IBM SPSS Version 30 (Build 172) confirmed significant agreement ($\kappa = 0.739$, $p = 0.007$). Consensus discussions were conducted by the authors to resolve discrepancies and ensure compliance with the PI framework (Tracy, 2013).

To evaluate students' critical thinking level while using ChatGPT, each chat transcript was coded (Table A1). Each code represents a participant's cognitive stage (Castellanos-Reyes et al., 2025), from one (triggering only) to all four in a cycle. Codes were based on stage presence and advancement, not frequency: Code 1 (25%), students demonstrated no critical thinking with only one stage; Code 2 (50%), low critical thinking with two stages; Code 3 (75%), moderate critical thinking with three stages; and Code 4 (100%), full interaction across all four PICP model stages. These codes matched "no critical thinking" to "critical thinker" levels.

4.3.3. Semi-Structured Interviews

To gain deeper insight into how GenAI (ChatGPT) influenced students' critical thinking, we conducted 45 min semi-structured interviews with two of the six participants. Both had already worked with the four stages of PI's cognitive presence, triggering events, exploration, integration, and resolution, as a structured guide while using ChatGPT during a two-week lesson planning task for their practicum. The interview protocol was designed

to investigate critical thinking in relation to the PICP model developed by Garrison et al. (2001). Each interview was audio recorded, transcribed verbatim, and analyzed using Braun and Clarke's (2006) hybrid thematic approach, which allowed us to move between deductive and inductive ways of making sense of the data. In the deductive phase, coding was guided by the dimensions of the PI's cognitive presence stages. The first author conducted the initial coding, and the third and fourth authors reviewed the process to ensure consistent application of the a priori coding scheme based on the PICP model.

In the inductive phase, attention shifted toward what emerged directly from the participants' own experiences of working with ChatGPT. The first author drafted initial codes, and the third author reviewed them carefully. Discussions facilitated the resolution of phrasing issues and the attainment of coding consensus. Subsequently, five themes were improved by the constant comparative method (Corbin & Strauss, 2014), which enabled the identification of patterns, categories, and coherence.

To enhance reliability across both phases, several strategies were employed. By bracketing direct quotations in the coding and theme analysis, meaning was maintained while admitting and capturing potential biases or opinions (Gearing, 2004). Reflexive memos were preserved during the coding process to monitor biases, document insights, and facilitate transparent decision-making (Creswell & Miller, 2000; Tracy, 2013). Collectively, these methodologies enhanced the credibility and reliability of the results.

5. Results

5.1. Students' Perceptions of the Impact of Generative AI on Their Critical Thinking (Q1)

Descriptive statistics, including the mean and standard deviation, were computed to address the first research question through quantitative data analysis using IBM SPSS software. The survey analysis encompassed 12 items measured on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). Table 4 illustrates the corresponding mean (M) and standard deviation (SD). To interpret agreement levels, the following scale was used: 1.00–1.80 = strongly disagree, 1.81–2.60 = disagree, 2.61–3.40 = neutral, 3.41–4.20 = agree, and 4.21–5.00 = strongly agree.

Table 4. Participants' perceptions of GenAI impact on their critical thinking.

Critical Thinking Skills	N	Minimum	Maximum	Mean	Std. Deviation
Triggering events	40	1	5	3.6	1.03
Exploration	40	1	5	3.7	0.97
Integration	40	1	5	3.6	0.99
Resolution	40	1	5	3.3	0.79
Average	40			3.5	1.1

Upon examining the survey data, which demonstrated strong internal reliability (Cronbach's alpha = 0.95), students agreed that GenAI could enhance their critical thinking, endorsing 8 out of the 12 statements, with the strongest agreement appearing in the exploration and integration phases of the cognitive presence framework.

Among the four cognitive presence dimensions, the exploration phase received the highest level of agreement (M = 3.70, SD = 0.97), indicating that students perceived ChatGPT as particularly useful for stimulating them to ask follow-up questions, brainstorming, and refining their understanding of content. Additionally, the triggering event and integration phases followed closely, each with a mean score of 3.60 (SD = 1.03 and SD = 0.99, respectively). These findings indicate a consistent perception that GenAI stimulates cu-

riosity and a sense of puzzlement, while also facilitating the synthesis and evaluation of information across conceptual boundaries.

Conversely, student responses concerning the resolution phase exhibited a neutral degree of agreement (M = 3.30, SD = 0.79), below the agreement level. This indicates their hesitant or inaccurate view of GenAI's capacity to assist with higher-order tasks such as applying knowledge, drawing conclusions, or solving real-world problems independently (see Figure 3).

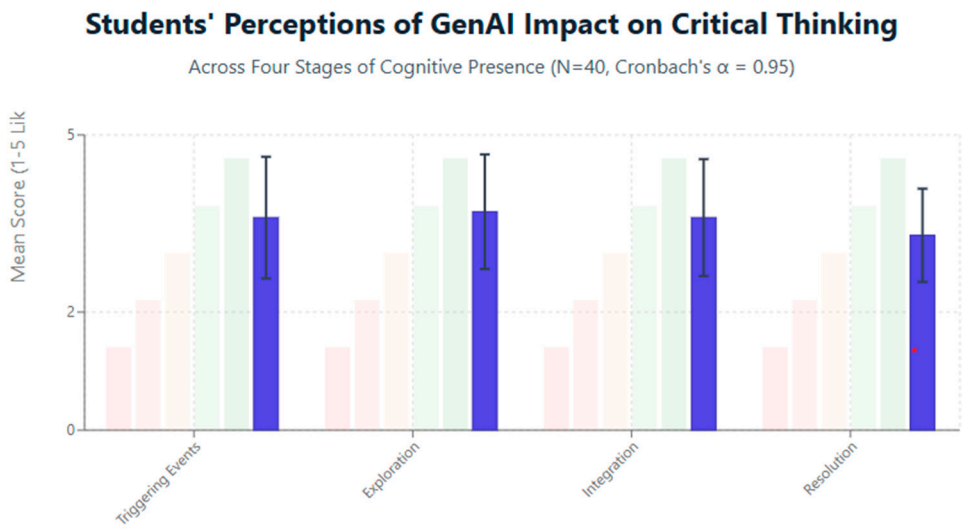

Figure 3. Survey results of students' self-reported critical thinking survey across the four stages of the PICP.

As a result, the data illustrate a clear pattern: Students viewed GenAI, particularly tools like ChatGPT, as a valuable cognitive partner during the earlier and intermediate phases of critical thinking, with uncertainty on whether to accept the final conclusion of their generated content in their real life or not.

5.2. The Extent to Which Students Demonstrate Critical Thinking While Using GenAI (Q2)

We coded each student's ChatGPT dialogue according to the priori coding scheme of cognitive presence (CP), (triggering, exploration, integration, resolution) (Garrison et al., 2001). This found two clear GenAI interaction patterns or modes, as indicated by Ouyang and Jiao (2021). In the first, a passive, AI-tutoring/directed pattern, students asked only one or two levels of questions and did not advance to deeper inquiry. In the second, a collaborative AI-supported mode, students engaged in multi-step dialogues, reaching the integration phase of critical thinking. Notably, no participant completed all four phases (Code 4) of the inquiry cycle (see Figure 4). Table 4 (above) shows that three of the six students were coded at Level 3 (75%, moderate CT), and the other three at Levels 1–2 (≤50%, no/low CT). In summary, half the participants showed only minimal cognitive presence in their GenAI interactions (Codes 1–2), and half showed moderate presence (Code 3); none achieved the full critical thinking cycle (Code 4).

5.2.1. Passive AI-Directed Interactions (Code 1–2; No/Low Critical Thinking)

In Chat 1, Kory posed seven straightforward, one-shot prompts, such as, "*What is a curriculum?*" and "*Provide a lesson plan for 5th grade*." All of her queries were triggering questions, resulting in Code 1 (25%, indicating a lack of critical thinking). No follow-up questions regarding exploration or attempts to connect ideas for integration were made. She accepted the responses from ChatGPT under the assumption that they were accurate and valid.

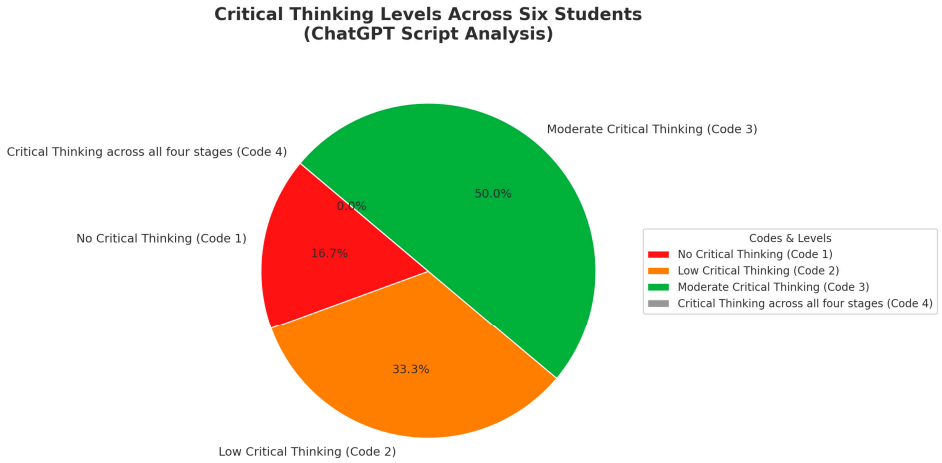

Figure 4. Distribution of critical thinking levels (Codes 1–4) across six students based on ChatGPT script analysis.

Additionally, in Chat 2, Mike posed seven questions, beginning with fundamental requests such as, "Check the syntax of this text" and "*Check the text for eloquent writing based on the title of the article*." The result placed 43% of the questions in the triggering stage and 57% in the exploration stage, with no instances of integration. This resulted in Code 2 (50%, low CT). She repeatedly demanded that ChatGPT rephrase or reword text, rather than commenting on or analyzing views.

Additionally, Riley in Chat 3 posed 15 unrelated questions (e.g., *"Can you explain curriculum theory in simple terms?"*) and terminology (e.g., *"What does communiqués mean?"*), with 66.67% of her inquiries classified as triggering and 33.33% as exploration. Nonetheless, none of her questions advanced to the level of integration; there was a lack of evidence for connecting ideas, evaluating information, or applying it to real-world situations. Her dialogue showed insufficient synthesis and application attempts, leading to a classification of Code 2 (50%, low CT).

In all three cases, the students used ChatGPT as a tutor, fact supplier, or *"answer machine."* Their exchanges were mostly solitary inquiries without iteration or thought. The GenAI responses were not linked to arguments, knowledge, or real-world situations. They only engaged in the triggering and exploration stages, demonstrating weak critical thinking.

5.2.2. Collaborative AI-Supported Interaction (Code 3; Moderate Critical Thinking)

Conversely, in Chat 4, Kate posed 31 inquiries covering many subjects, such as *bilingual education, narrative inquiry, and Hmong education*. Her inquiries encompassed a proportional allocation of triggering (45.10%), exploration (45%), and integration (12.90%). She progressed from fundamental definitions to the comparison and evaluation of concepts, such as inquiring about "*methodologies in narrative inquiry and contextualizing Hmong student achievement*." Her interaction exhibited involvement with ChatGPT through three stages of critical thinking (CT), categorizing her as Code 3 (75%, moderate critical thinking).

In Chat 5, Emily posed 35 questions that centered on motivation, classroom culture, folktales, and lesson planning, showcasing advancement through various cognitive stages. Her questions, like, *"What is a legend?" Could you provide me with some examples of legends? Do legends always have their roots in real individuals? Are folklore and folktales the same or different? Would a folktale be both traditional literature and folklore? Would traditional literature and folklore be the same thing then? What are some examples of folktales? Isn't Cinderella a fairytale? I have read that before!"* spanned triggering (37%), exploration (43%), and integration (20%). Consequently, no clear conclusions or practical applications were clear in her interaction. She initiated her conversation with a sense of puzzlement and posed further inquiries

to enhance her conceptual understanding. She integrated ChatGPT's responses with her existing knowledge, particularly when she scrutinized the AI's classification of Cinderella. This indicates how she was actively engaging in a collaborative discourse with ChatGPT, classifying her as Code 3 (75%, moderate critical thinking).

Likewise, in Chat 6, Robe posed 20 questions concerning reading instruction and dyslexia. The distribution of his questions indicated 30% triggering, 50% exploration, and 20% integration, with no evidence of resolution present. She advanced from inquiring, *"What is the process of how children learn to read?"* to *"How does dyslexia affect this?"* and ultimately to *"What research papers offer valuable information on testing and dyslexia?" Some references are inaccurate; could you please provide the correct ones?* The interconnected prompts, demonstrating authentic inquiry and assessment, led to Code 3 (75%, moderate critical thinking).

Together, these three students used ChatGPT to have what you might call a "*collaborative dialogue*." They asked follow-up questions that were built on previous replies, tried to bring together and make sense of information, and hinted at possible uses, but none of them made it to the final stage of resolution. Their use of GenAI is similar to a joint inquiry model, in which students worked together to find answers to questions by improving their questions and thinking about the answers over and over again. Each of these transcripts spanned three cognitive stages, indicating a meaningful engagement with critical thinking beyond superficial Q&A (See Table A2).

5.3. The Impact of Guided GenAI ChatGPT Use on the Development of Critical Thinking (Q3)

To address the third research question, students' perceptions of the impact of guided GenAI ChatGPT use on the development of their critical thinking, a thematic analysis of the two interviews with Robe and Emily was conducted. Five major themes were identified through the analysis, reflecting the dynamic nature of students' engagement with ChatGPT under structured pedagogical conditions (see Table A3 and Figure 5).

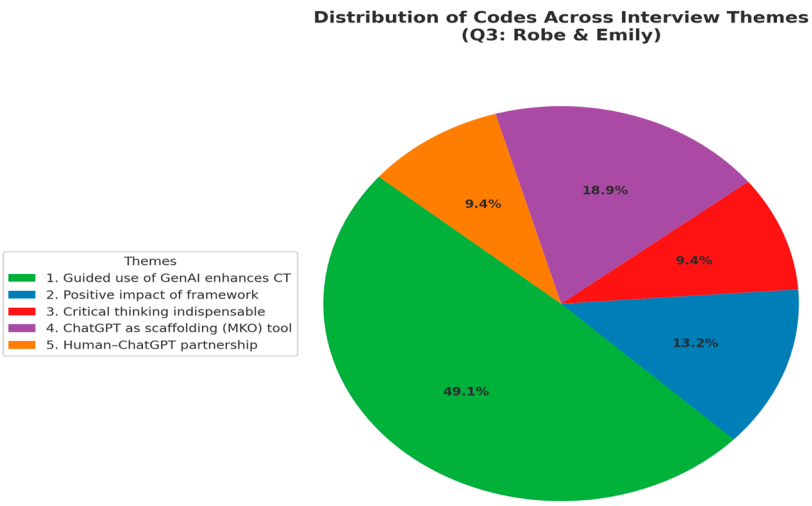

Figure 5. Interview themes and distribution of codes (Q3: Guided GenAI use and critical thinking).

5.3.1. Guided Use of GenAI Was Associated with Reported Improvements in Critical Thinking

As for triggering events, the first stage of cognitive presence, students reported that ChatGPT initially sparked curiosity and broad inquiry. Participants described a sense of puzzlement that led them to test the system. For example, Robe recalled that *"the first time I was using ChatGPT... I felt like I should ask the most broad questions possible in order to get answers that I would want."* Robe even framed his initial prompts as experiments to compare his own understanding with the AI's: *"I wanted to see if I had a full understanding of it, was*

my understanding correct? Did ChatGPT have the same understanding as me? I could then realize that it was capable of answering questions that were very advanced."

Emily, likewise, turned to ChatGPT when encountering unfamiliar content: *"I used ChatGPT a lot in my multicultural education classes to help me kind of understand topics that I hadn't learned about prior to."* In both cases, the AI raised initial questions and made students reflect on what they did and did not know. *"I would go to my assignment and look at what I had to do. And I would think, okay, where do I not understand a certain section. Sometimes in the books we are given, it can be a lot of words that I don't understand, so I used ChatGPT in that way."* These comments align with the triggering events definition, a moment of puzzlement and curiosity, as students sought to use ChatGPT to find solutions or clarify confusion. Thus, ChatGPT capabilities stimulate students to ask and seek information—the first step of critical thinking: to have the curiosity to discover.

In the second phase, students dug deeper into topics by refining their questions and engaging in iterative dialogue with ChatGPT by asking follow-up questions and requesting clarification by asking for examples. Robe described how she moved from broad inquiries (e.g., about dyslexia and testing) to very specific follow-ups: *"Once I finally got down to the point of asking, How does dyslexia affect testing?, I was getting pretty specific answers, and it was actually helping out a lot. It gave me very detailed step-by-step stages, different important resources that children use. Then I dug a little deeper and asked, How does dyslexia affect their reading? Even in the case where I wanted to continue my learning, I could ask ChatGPT, What are good examples of research papers that study this topic? and it could give me better answers than I could find on Google Scholar or through the library catalog. They were all very connected."*

As noted, Robe indicated that ChatGPT gave detailed, step-by-step explanations and resources, which prompted her to dig a little deeper: *"I asked this question, this question gave me this answer, and from this answer I can properly know to ask this next question."* Similarly, Emily reported that ChatGPT became more conversational and exploratory over time. Instead of doing simple Google searches, she used it as an interactive conversation tool to map out assignment tasks: asking for first-week class procedures, then breaking them down into daily schedules, and then seeking definitions: *"Instead of going to Google to look up websites and articles, it was more of a conversation tool for me. I would ask what are some first week procedures... then go deeper into a grade level now... break it into a time schedule... I want to go further in depth of what is intrinsic motivation... then I can go into my assignment and write out... some of the ChatGPT ideas and incorporate that into my actual work. It helps me kind of think of what information would be useful to use in my work, and what information I need to look for."*

As seen in their quotations, this iterative questioning encouraged them to analyze what information was needed and how to frame each query. Both students indicated that specificity and follow-up questions improved the usefulness of the responses. These behaviors reflect deep exploration: students brainstormed ideas, requested examples, and continuously adjusted their prompts to extract more useful information from ChatGPT.

In the third phase, students compared ChatGPT's responses with their own background knowledge and other resources, deciding what to accept or adapt. Robe reported that ChatGPT's answers closely matched the curriculum and her prior understanding: *"Not only did it help me understand the project better, but it actually helped me better prepare for what I'm going to be doing in the future if I want to be teaching sixth-grade social studies. It gave me the curriculum overview. And I could understand where the topics that they were assigning me now connected to the broader topics. I had to break down every question and look into the answer for every single one in order to get the response I was hoping for. When I asked it to give me references for research papers, I discovered the citations were fake, so I asked it to give me the correct citations after I provided it with the right references."*

As you see, what Robe indicated is an example of how GenAI helped her integrate new information into her existing schematic learning and how her way of guiding it by checking the references made the interaction active and the results reliable. Emily likewise found that ChatGPT both affirmed what she already knew and introduced new ideas. She used her own ideas as a baseline and then asked, *"I kind of used my previous ideas for certain work, and then I would think, okay, I have a good baseline. It helped me understand parts of my background knowledge... and it would come up with a whole different form of ideas and prior ideas I had thought of. ChatGPT gave a lot of answers... Okay, I kind of already knew of this idea, but what had I not heard of yet? It was nice to get new ideas, and it would explain further some of my background knowledge, or prior information."*

In this way, ChatGPT *"gave a lot of answers"* and Emily picked out novel points to incorporate based on comparing and contrasting to what she already knew. Both participants also reflected that they often had to deliberately weigh and synthesize ChatGPT's output: Emily noted that, while some suggestions overlapped with her background knowledge, others filled gaps she had not considered. In summary, during integration, students actively negotiated meaning between AI and their own schemas: They accepted some AI-generated ideas, rejected others, and combined elements to refine their understanding.

In the final phase, Robe and Emily applied what they learned to produce actual work or final solutions, using ChatGPT's output as a springboard. Neither student simply copied AI text into their assignments; instead, they repurposed and transformed it. Robe insisted that *"I never wanted to directly pull a quote from ChatGPT and put it directly in my project... So it gave me a good starting point that I could assign... or make it the theme of one of my slides* (e.g., *testing accommodations for dyslexic students),"* which she then incorporated into slide themes and planning. Working at a testing center for dyslexic students herself, she said ChatGPT *"helped give me the different parts that I need to be focusing on… it gave me a good starting point that I could… make it the theme of one of my slides that I managed to apply in my work, a testing facility helping students with dyslexia take tests."*

Additionally, Emily similarly edited the AI's *"robotic"* language into her personal voice: *"I took a lot of my answers from ChatGPT, and I would configure them to be my own ideas… I would transform their words, because sometimes… it would be a little robotic… and build off of it."* She also used ChatGPT to format components of assignments. She mentioned, *"I had to make a communication log for my students' parents. I went to ChatGPT and just asked it... and formatted it into my own log. I would transform their words to add my voice. It helped a lot with the different perspectives before I complete it and use it in my class... it was nice to go to ChatGPT for further clarification on certain topics."*

In both cases, ChatGPT acted like a collaborative assistant or the more knowledgeable other, scaffolding their learning: It provided structure and content that the students then contextualized and refined. By resolution, each student had organized the AI-derived ideas to create a coherent final product or plan, effectively solving their problem with the aid of both the AI and their own critical insights.

5.3.2. Positive Impact of the Used Framework on GenAI Use

After moving through all four stages of the cognitive presence framework, students reported a striking shift in how they saw ChatGPT's role. Initially, both had viewed it as a potential *"quick fix,"* but using it within the cognitive presence framework reframed it as a thinking partner. Robe explicitly said she came to see ChatGPT *"not just… as giving me answers; but as pushing me to ask better questions, organize my ideas, and come up with real, practical ways to support students."* She now considers GenAI a *"thinking partner"* that sharpens her reasoning without replacing her creativity or empathy: *"As a future educator, I really see AI now as more of a thinking partner. It's not replacing my creativity or empathy. It's*

just helping me sharpen my thinking, organize ideas, and even rehearse lessons. I definitely think it should be in the education system. But in order to be effective in an educational sense, you do have to apply the cognitive presence as a critical thinking stages."

Emily mirrored Robe's reflections on the use of the CP stages. She noted, *"I think CP stages helps evoke critical thinking skills... you might have to have some critical thinking to use ChatGPT. So instead of just copying what it said, I used it more like a collaborator... It helped me think through different ways to communicate clearly and professionally. The PICP Critical Thinking model helps to know that it's not about just getting an answer, it's about learning how to shape the right questions and then figure out how to apply what you find. ChatGPT worked with me, as a kind of planning partner... Especially where tone and trust really matter."* Both students emphasized that effective use of ChatGPT required active critical engagement. As Robe put it, *"you do have to apply critical thinking skills... it wasn't just about getting an answer."* Similarly, Emily said the process *"helped me develop my thought process... not just getting an answer but learning how to shape the right questions."* In sum, completing the four stages convinced them that ChatGPT could stimulate deeper thinking.

5.3.3. Critical Thinking Is Indispensable in Using GenAI

Across all stages, students repeatedly underscored that students must bring their own critical thinking to the table. Effective use of ChatGPT requires reflective judgment, evaluation, and the ability to analyze and apply content independently. For instance, Robe noted that to make the tool educationally effective, *"you do have to apply critical thinking skills,"* and that the framework's emphasis on questioning was vital. Emily observed that the exercise *"evokes critical thinking skills... it was trial and error... you have to get comfortable"* with prompting. Both mentioned having to break down questions and answers to extract what was truly important. These reflections illustrate that the interviews themselves instilled a mindset of skeptical inquiry: The students repeatedly said they learned to *"look deeper," "analyze,"* and *"find what was specifically valuable"* in ChatGPT's responses. This theme of cautious scrutiny recurred: Even as they grew more positive about the tool, they stressed that learning from it required constant self-questioning and deliberation.

5.3.4. ChatGPT as a Scaffolding More Knowledgeable Other (MKO) Educational Tool

Both participants viewed ChatGPT as a practical educational support tool, especially for writing, note-taking, idea synthesis, and accessing resources. Robe called it *"a great note-taking tool... a great way of synthesizing thoughts into easier to manage sections."* Emily agreed, noting that ChatGPT *"helped me work a lot better and quicker than I would have if I'd gone to Google by breaking everything down into much easier to manage groups of information."* They reported using it to brainstorm quiz questions and lesson ideas (*"quizzes and tests, questioning, exit tickets"*), and even to find academic resources that Google could not easily locate. However, they emphasized that the AI was valuable because it encouraged introspection and metacognition, an impact they became aware of only after they deliberately scrutinized and arranged the AI's output, as some of it was misinformation and fake. Robe noted that *"ChatGPT gave a lot of information quickly,"* but *"I had to go through and make sure it was actually accurate, and some references were fake."* Emily similarly admitted she was *"a little wary of exactly where it comes from or if there was anything incorrect in it."* While they appreciated ChatGPT's time-saving convenience and broad output, both students recognized its limitations and made a deliberate effort to verify and refine its responses, an act that, in turn, fostered introspection and metacognitive awareness.

5.3.5. The Nature of the Human–ChatGPT Partnership

Finally, students described their relationship with ChatGPT as evolving from an impersonal tool to a personalized collaborative partner. They emphasized the conversational,

adaptive nature of the interaction. For them, it represents the more knowledgeable other. Robe said, *"I came to see AI as a genuine buddy"* in educational planning. Emily noted, *"I would definitely say, collaborator... I took the answers, and I would transform them and go deeper into it with ChatGPT. It helped me shift perspectives too, like thinking not just as a teacher, but also thinking about what a parent might be feeling."* Importantly, they felt the AI *"listened"* to their needs: As they asked more informed questions, ChatGPT gave more targeted answers, making the exchange feel increasingly reciprocal. Also, Robe noted, *"There's a difference between getting an answer to a question... versus asking a question and analyzing every part of it."* Initially, both thought ChatGPT was *"just a quick fix,"* but over time they *"trusted"* it more as an intelligent partner that could accommodate their individual learning styles. In short, ChatGPT came to be seen not as a static database but as a dynamic, responsive presence in the learning process.

6. Discussion

The study results indicated a multi-dimensional understanding of how students perceive and engage with GenAI in ways that influence their critical thinking. Moving beyond surface-level agreement in the survey, the chat histories and interviews demonstrate how students apply critical thinking in light of the four cognitive presence stages: triggering, exploration, integration, and resolution, when interacting with ChatGPT. In this discussion, we examine the depth of students' cognitive engagement, the nature of their collaboration with GenAI, and the pedagogical implications of their evolving perceptions. Using this perspective, we contend that the value of GenAI is found in its design, which encourages thoughtful, iterative, and context-aware thinking, rather than in the way content is delivered.

6.1. What Do the Survey Responses Indicate?

According to the quantitative survey results (Q1), students thought ChatGPT was helpful for early-stage critical thinking, fostering curiosity, idea formation, and integration; however, they were not as sure of its significance in the last, resolution/breaking the complexity stage. According to Table A2, students strongly agreed that ChatGPT supports triggering events (M = 3.60) that raise more inquiries as well as facilitate concept exploration (M = 3.70) and integration (M = 3.60). On the other hand, their responses regarding how well ChatGPT helps them break down complexities and makes it easier to draw conclusions (Resolution) were neutral (M = 3.30).

Students identified ChatGPT as a useful tool for brainstorming and cognitive engagement, yet they remained prudent in regarding its outputs as definitive solutions. This pattern aligns with the findings of Miller et al. (2025), which indicated that ChatGPT can alleviate cognitive load and enhance critical thinking by managing complex tasks, and Guo and Lee's (2023) findings that students perceived ChatGPT as a source of "diverse perspectives" that prompted them to reconsider their existing thought processes, which is evident in our elevated exploration and integration scores. However, they encountered difficulties in validating the information generated by AI. Survey results indicated that ChatGPT could enhance CT as it facilitates inquiry and synthesizes ideas; however, its effectiveness in drawing conclusions and applying information is still unclear. The findings by Wang and Fan (2025) indicate that the beneficial impact of ChatGPT on higher-order thinking is achieved solely through the implementation of instructional scaffolds. Without structured guidance, AI may assist with idea generation and brainstorming but may not independently lead to final conclusions. Therefore, educators should train students in precise, open-ended, and iterative prompting and critical evaluation so that ChatGPT becomes an active partner in learning rather than a shortcut.

6.2. Two Interaction Pathways with ChatGPT: AI Tutoring vs. Collaborative Personalized Support

Students' mixed confidence suggests that without such scaffolds, they view ChatGPT more as a curiosity stimulator than a reliable tutor: It sparks questions and clarifies concepts (triggering/exploration) and helps them connect ideas (integration), but they remain doubtful about accepting AI-generated solutions (resolution). These survey attitudes matched the chat transcript analysis (Q2). We found two distinct interaction patterns. In the passive AI-directed paradigm (3 of 6 students), learners treated ChatGPT as a one-way information source. They asked only straightforward, one-off questions (triggering), accepted the answers at face value, and did not pursue follow-ups or synthesize responses. These students' engagement stopped at triggering or, at best, low-level exploration, yielding Code 1–2 (no/low CT). This is consistent with reports that many students offload thinking to AI.

For example, Costa et al. (2024) discovered that ChatGPT users displayed metacognitive laziness or poor cognitive engagement, relying on ChatGPT to undertake the mental rigor of analysis and synthesis. Similarly, Larson et al. warn that GenAI's polished, confident outputs can "inhibit critical thinking" because students tend to accept them uncritically on the assumption that they are correct (Essien et al., 2024; Valova et al., 2024).

In the passive mode, students indeed treated the AI's responses as authoritative (e.g., asking straightforward definitional prompts without clarification), a behavior consistent with the cautionary viewpoint of Larson et al. (2024). However, when combined with appropriate scaffolding, ChatGPT can enhance higher-order thinking, according to a recent meta-analysis (Wang & Fan, 2025). Most importantly, they emphasize that teachers should use ChatGPT as a guided tutor or learning partner and provide suitable scaffolds or educational frameworks (e.g., Bloom's taxonomy).

In contrast, the collaborative AI-supported group that exhibited moderate critical thinking approached this ideal. These learners engaged in iterative, multi-step dialogues. They initiated with sense-making inquiries, subsequently posed several precise follow-up questions, compared AI responses to their pre-existing knowledge, and progressed towards synthesis (integration), although none achieved complete resolution. This became evident when Emily challenged ChatGPT's categorization of "Cinderella," while Robe expressed skepticism regarding fabricated citations and employed AI-generated curriculum insights to improve her practical lesson planning. The findings of Suriano et al. (2025) indicate that engagement with GenAI can improve students' critical thinking skills, contingent upon the design of learning experiences that promote active participation and a deeper comprehension of AI-generated responses.

Two modes of interaction emerged from the chat transcript analysis: AI tutoring (also referred to as AI-directed) and AI collaboration (or AI-supported). In the former, ChatGPT functioned as a one-way knowledge source, simply supplying answers, and critical thinking remained low. In the latter, students treated GenAI as an interactive, collaborative partner, more akin to a guided peer. This distinction aligns with other studies of AI in education suggesting that ChatGPT can serve as a collaborative assistant in writing tasks (Luther et al., 2024).

The moderate critical thinking participants displayed a similar dynamic at a higher level: They engaged in mini-dialogues that required reading GenAI responses and then building on them after comparing and contrasting the content with their social schema. Nevertheless, none achieved complete resolution or implemented their knowledge of GenAI in practical scenarios. This indicates that while viewing GenAI as a collaborator may promote reflective questioning, more guidance or more intricate problems are required to obtain comprehensive critical thinking results. The conversation script corroborates the survey's suggestion that ChatGPT fosters curiosity, exploration, and integration stages, although it reveals a deficiency at the resolution stage that students fail to address inde-

pendently. As a result, to harness GenAI for critical thinking, students need guidance and scaffolds. When thoughtfully integrated, as suggested by prior research, ChatGPT can become a catalyst for analysis and synthesis (Wang & Fan, 2025); without scaffolds, it risks becoming a shortcut around deep critical thinking (Kestin et al., 2025; Larson et al., 2024).

6.3. Filling the Critical Thinking Gap Through Structured ChatGPT Use

The interview findings (Q3) illustrate how this gap can be filled when students follow a guided and structured inquiry framework while using ChatGPT. Their perceptions shifted dramatically, from viewing it as a mere source of knowledge; to accepting it as the more knowledgeable other and the collaborative cognitive partner that can stimulate their thinking, much like the evolution from traditional learning, where students perceived teachers as the sole source of knowledge; to more contemporary learning approaches in which learners engage in specific processes to learn in an active and engaging way.

Firstly, during the triggering and exploration phases, students engaged with ChatGPT extensively, refining their uncertain inquiries over time. This finding corroborates literature suggestions that deliberate GenAI prompting is essential for cognitive engagement, and that when students learn to formulate precise, open-ended, and iterative prompts, they can leverage AI to enhance their cognitive involvement (Walter, 2024). More specifically, Robe and Emily explicitly reported that refining their prompts (e.g., narrowing broad questions about dyslexia into specific sub-questions and refining the ChatGPT responses by asking and incorporating that into their actual work to think of what information would be useful to use in their work) led to richer answers. This process of iterative questioning supports critical thinking by forcing students to break problems down into sub-tasks and evaluate each AI response, as indicated by Wang and Fan (2025).

Secondly, in the integration and resolution phases, both students demonstrated critical involvement by juxtaposing ChatGPT's output with their prior knowledge and customizing the content to suit their tasks. Instead of replicating comments, Robe and Emily modified the wording and structure of AI-generated material to reflect their own voices. This affirmed that they utilized ChatGPT not as a mere shortcut but as a cognitive collaborator. Their conduct corresponds with Essel et al. (2024), who discovered that flipped classrooms fostering student accountability in assessing AI content enhanced critical thinking abilities. Likewise, the participants observed that ChatGPT assisted them in establishing curricular linkages, including *"curriculum overview... broader topics,"* and applying them to practical situations such as instructional planning and assessment modifications. These findings align with Wang and Fan's (2025) results that ChatGPT can significantly enhance higher-order thinking when used in a structured way. Remarkably, they successfully completed all four stages of cognitive presence, including the resolution step, which was absent from their conversation scripts. This indicates that the structured utilization of GenAI facilitated their attainment of this last stage and demonstrated critical thinking when interacting with GenAI ChatGPT.

Additionally, they redefined ChatGPT as a *"cognitive collaborator."* They mentioned that ChatGPT stimulated them to ask more in-depth inquiries and helped them organize their views. Results indicate that students' use of the framework facilitated and changed their perceptions. Students transitioned from perceiving ChatGPT as a mere shortcut to utilizing it as an interactive collaborator that facilitated their research, a trend corroborated by Suriano et al. (2025), who discovered that active contact with AI chatbots enhances critical thinking growth.

The PICP model used also seemed to address concerns about passive GenAI use. Participants described how, during the integration phase, they critically evaluated GenAI content to avoid hallucinations and ensure integrity. As Robe recounted, *"When I asked*

it to give me references for research papers, I discovered the citations were fake, so I asked it to give me the correct citations after I provided it with the right references." Similarly, Emily noted, *"You might have to have some critical thinking to use ChatGPT. So instead of just copying what it said, I used it more like a collaborator... It helped me think through different ways to communicate clearly and professionally."* Additionally, Robe mentioned, *"It wasn't just giving me answers; it was pushing me to ask better questions,"* while both students reflected on how the process demanded more *"evaluation and analyzation... than they were prepared for."*

In terms of ChatGPT as an educational tool, students were generally satisfied with its practical benefits. They noted that ChatGPT quickly synthesized information that would have taken much longer to find via Google. As Emily stated, *"Instead of going to Google to look up websites and articles, it was more of a conversation tool for me."* They even used it for resource-finding (e.g., research papers) that would be difficult to locate manually. Cooper (2023) affirmed this viewpoint, mentioning that ChatGPT was used as a research tool for assistance with editing and to experiment with making the research narrative clearer. While Robe noted the problem of fake sources, she was able to guide ChatGPT to locate correct ones by using the structured framework. These cautious reflections parallel those of Blahopoulou and Ortiz-Bonnin (2025), who observed that students worry about academic integrity and information accuracy when relying on AI.

Both Robe and Emily consistently emphasized this vigilance. They used ChatGPT to synthesize and brainstorm, but always fact-checked and refined its outputs. As Robe explained, *"Like something people used to skip out on doing the real thinking. But after I started using the Critical Thinking Framework with it, I noticed it really helped me think more deeply. It wasn't just giving me answers; it was pushing me to ask better questions, organize my ideas, and come up with real, practical ways to support students. So ChatGPT will never replace us, it complements and enhances our critical thinking."*

In conclusion, both participants highlighted that the use of ChatGPT required critical thinking. According to Robe, ChatGPT presents dual aspects: *"It necessitates critical thinking skills... one may need to employ critical thinking to utilize it."* Olivier and Weilbach (2024) contend that students' perceptions of GenAI are influenced by their instruction on its use as *"a cognitive tool rather than a replacement for independent reasoning."* Participants exemplified this method through the analysis of AI drafts, source verification, and critical engagement with the information. This corresponds with an expanding collection of studies suggesting that AI may either impede or facilitate critical thinking based on its application (Suriano et al., 2025; Walter, 2024; Zhou et al., 2024), a notion emphasized by Robe's reflection.

7. Implications

7.1. How Can GenAI Be Positioned as a Thinking Partner?

The findings of the current research present important implications for educators and curriculum designers. First, they suggest that considering ChatGPT as merely an answer-generating tool is misguided. Instead, students must be guided to use it critically and responsibly, as Ritz et al. (2024) suggests that unguided utilization of ChatGPT reduces cognitive engagement. They advocate for pedagogical approaches that promote critical and active engagement with AI-generated material. When integrated within a systematic education that emphasizes self-reflection, metacognition (Garrison, 2022), and intellectual autonomy, its capacity to cultivate authentic critical thinking is completely actualized.

7.2. GenAI Literacy and Structured Use of GenAI

This research confirmed that without structure, student interaction with GenAI tends to adopt a passive, transactional approach to information retrieval, which may negatively impact cognitive effort and encourage superficial learning, as also noted by Ritz et al.

(2024). Conversely, when students applied the Practical Inquiry model of cognitive presence, their interactions transformed into a more collaborative, dialogic process characterized by iterative inquiry and analytical participation. This distinction highlights an essential responsibility for educators: to enable the use of AI through inquiry-based frameworks that convert students from passive recipients of knowledge to active participants in discourse (Wang & Fan, 2025). Moreover, fostering GenAI literacy and a mindset of critical skepticism is crucial for equipping learners to engage with GenAI proficiently (Walter, 2024). Considering that GenAI outputs may be generalized, incorrect, or even faked, learners must receive explicit training to analyze AI-generated content, recognize logical inconsistencies, and meticulously corroborate assertions using reputable sources (Miller et al., 2025).

7.3. Adapt Current Assessment Methods to GenAI

The current pedagogical shift, driven by the widespread use of GenAI and its provision of vast amounts of information, much of which may be unreliable, necessitates a corresponding transformation in assessment practices. Traditional evaluation rubrics that emphasize response accuracy alone may inadvertently promote the instrumental and uncritical application of ChatGPT. Greater emphasis should be directed toward the learning process rather than exclusively on learning outcomes. To foster profound learning, assessment models must be reconfigured to highlight intellectual engagement, particularly the development of incisive inquiries, the refinement of critiques aimed at AI-generated content, and the ability to integrate disparate ideas into a cohesive conceptual framework. This approach emphasizes the importance of involving students in challenging tasks that foster advanced cognitive skills and intricate problem-solving, leading to emotional involvement, observable actions, and significant outcomes. As Van den Berg and Du Plessis (2023) assert, promoting epistemic curiosity is essential for positioning GenAI as a credible foundation for the advancement of critical thinking.

7.4. Toward AI-Mediated Inquiry Grounded in Human Oversight

Ultimately, these findings necessitate a paradigm shift from rigorous standards to the establishment of AI-mediated inquiry in educational environments in which human agency is the primary influence rather than automation. This involves equipping students to adeptly navigate the limitations of AI's functionalities and cultivating a robust internal framework for ethical deliberation and credibility assessment (Guleria et al., 2023; Tam et al., 2023; Wójcik et al., 2023). By situating GenAI within a framework like PICP (refer to Figure 2), educators may facilitate the ethical incorporation of technology while fostering learners' intellectual independence, guaranteeing that students retain authority over their own knowledge acquisition. The objective is to not merely utilize AI but to engage with it as a reflective collaborator in a process that enhances the distinctly human capacity for advanced cognitive reasoning.

8. Conclusions

These preliminary findings suggest that the impact of GenAI on students' critical thinking may be influenced not by the technology itself but by how learners interact with it. Survey and ChatGPT script results indicated that students agreed ChatGPT enhanced the initial three stages of critical thinking by assisting them in generating, exploring, and integrating ideas with their prior knowledge; nevertheless, they expressed uncertainty about reaching resolution without structured guidance.

The analysis of six ChatGPT scripts distinguished between passive users who accepted AI responses without question and more collaborative users who demonstrated moderate critical thinking. However, these students rarely moved beyond integration. The interview

results offered additional insight, revealing that two participants progressed only when they were guided through the four stages of the PICP. It is noteworthy that no students were able to independently complete all four phases of CT, highlighting the necessity of structured instruction to enhance engagement in the resolution stage. Despite the interview findings being derived from a limited participant pool, which diminishes their generalizability, they demonstrate how structured instruction can facilitate deeper avenues of critical thinking. Students started to perceive GenAI ChatGPT not as an infallible authority but as a more knowledgeable counterpart, capable of fostering curiosity, synthesis, and critical analysis when utilized purposefully. As a result, GenAI does not replace human cognition but rather enhances existing cognitive abilities. In unstructured use, it has the potential to devolve into a heuristic shortcut that impairs critical thinking and comprehension. However, within regulated pedagogical settings, it can help scaffold metacognitive regulation and cognitive development. Its actual potential is not in automating cognitive labor but in enhancing inquiry as a socially situated process.

8.1. Future Directions

Future research should investigate how the long-term application of the PICP model aids learners in completing the inquiry cycle and improving their critical thinking skills while utilizing GenAI. Our short-term intervention showed that when students are guided and have some literacy in how to use GenAI, they reach the resolution stage.

Deeper and longer-term studies are needed to determine whether guided GenAI use enhances students' critical thinking and decreases reliance on surface-level outputs. Therefore, comparisons between structured and unstructured GenAI integration may clarify the influence of instructional design on the quality of students' cognitive processes.

Furthermore, analyzing student interactions with AI, especially the transition from passive acceptance to dialogic collaboration or AI empowerment, will augment the educational potential of GenAI. Further research should investigate the role of teacher facilitation and examine how students' ethical concerns affect their reluctance during the resolution phase.

Ultimately, future investigation should move beyond tool effectiveness to investigate how GenAI can be purposefully integrated as a reflective partner and a more knowledgeable other, one that challenges, supports, and extends the learner's critical thinking process without replacing it.

8.2. Limitations

The findings of this study should be considered in light of its methodological and demographic limitations. The study relied on a relatively small and demographically limited sample, all drawn from a single college of education, which constrains broader applicability. Dependence on self-reported data and a deductive coding methodology may have neglected emergent patterns. Using the Practical Inquiry Cognitive Presence Model for only two participants limits the generalizability of these findings and may have prevented deeper interpretive insights. Early 2023 data were collected using ChatGPT 3.5, which has since improved. The study did not assess learning or causality. These preliminary findings require further research using different samples and longitudinal methods.

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Abbreviations

The following abbreviations are used in this manuscript:

GenAI	Generative Artificial Intelligence
PI	Practical Inquiry
CP	Cognitive Presence
PICP	Practical Inquiry Cognitive Presence
CoI	Community of Inquiry
CT	Critical Thinking

Appendix A

Appendix A.1. ChatGPT Scripts

Table A1. ChatGPT script themes, code definitions, and occurrence rates.

PICP Critical Thinking Stage Definitions	Code Frequency	Level of Critical Thinking	Definition of Code
1. Triggering Events A sense of puzzlement that indicates a curiosity to ask a question or seek a solution for a problem. Then they start to ask GenAI a question to find an answer to the question or better solve a problem.	Code 1 (25%) One stage	No	Students demonstrate one cognitive stage only. Typically, a single, isolated question (one-shot or a single-level prompt). (triggering, exploration, integration, or resolution). This shows narrow engagement. This does not indicate the quality is low, but that the interaction lacks diversity in thinking.
2. Exploration Diving and digging deeper to search for more information, looking at different ideas, and trying to understand the problem better through asking follow-up questions, brainstorming, or asking for some examples.	Code 2 (50%) Two stages	Low	Students ask at least two related questions (two-shot) covering two distinct stages (e.g., triggering and exploration or integration and resolution). These stages may appear in any order and show an emerging effort to deepen or apply thinking, but without full synthesis.

Table A1. *Cont.*

PICP Critical Thinking Stage Definitions	Code Frequency	Level of Critical Thinking	Definition of Code
3. Integration Trying to make meaning by comparing the responses from GenAI with their background knowledge, social schema, and other external resources for validation, to decide what makes sense and what does not, and whether to accept it or adapt it by adding what they need to reach a final outcome from their collaborative discussion with GenAI.	Code 3 (75%) Three stages	Moderate	Students ask at least three logically connected shots or questions covering three stages, with visible effort to connect, synthesize, and reflect on AI input. This may include linking ChatGPT responses to prior knowledge, identifying contradictions, or preparing for application.
4. Resolution Students break down the complexity of the issue, problem, or idea to reach a conclusion or solution. They take everything they have learned and use it to organize their ideas, solve a problem, contextualize it, create something that fits the real situation, and apply it.	Code 4 (100%) Critical thinker The full cycle of four stages	Critical thinker	Students' multi-shot questions cover all four stages (triggering → exploration → integration → resolution) in their interaction through multi-level shot questions and clarifications. This represents a full critical thinking cycle, showing curiosity, depth, synthesis, and contextualized application of knowledge.

Appendix A.2. ChatGPT Script

Table A2. ChatGPT script analysis rubric (coding items and number of occurrences).

ChatGPT Script	Code	Number of Shots in Each Stage	Occurrence Ratio Scale with Critical Thinking Level
Chat 1: Kore asked 7 questions.	1. Triggering event—sense of puzzlement 2. Exploration—information exchange 3. Integration—connecting ideas 4. Resolution—applying new ideas	7 (100%) 0 (0%) 0 (0%) 0 (0%)	Code 1 (25%) No critical thinking
Chat 2: Mike asked 7 questions.	1. Triggering event—sense of puzzlement 2. Exploration—information exchange 3. Integration—connecting ideas 4. Resolution—applying new ideas	3 (43%) 4 (57%) 0 (0%) 0 (0%)	Code 2 (50%) Low level of critical thinking
Chat 3: Riley asked 15 questions.	1. Triggering event—sense of puzzlement. 2. Exploration—information exchange. 3. Integration—connecting ideas. 4. Resolution—applying new ideas.	10 (66.67%) 5 (33.33%) 0 (0%) 0 (0%)	Code 2 (50%) Low level of critical thinking
Chat 4: Kate asked 31 questions.	1. Triggering event—sense of puzzlement 2. Exploration—information exchange 3. Integration—connecting ideas 4. Resolution—applying new ideas	13 (42%) 14 (45.10%) 4 (12.90%) 0 (0%)	Code 3 (75%) Moderate level of critical thinking
Chat 5: Emily asked 35 questions.	1. Triggering event—sense of puzzlement 2. Exploration—information exchange 3. Integration—connecting ideas 4. Resolution—apply new ideas	13 (37%) 15 (43%) 7 (20%) 0 (0%)	Code 3 (75%) Moderate level of critical thinking
Chat 6: Robe asked 20 questions.	1. Triggering event—sense of puzzlement 2. Exploration—information exchange 3. Integration—connecting ideas 4. Resolution—applying new ideas	6 (30%) 10 (50%) 4 (20%) 0 (0%)	Code 3 (75%) Moderate level of critical thinking

Appendix B. Interview

Appendix B.1. Interview Questions

Triggering Events (Identifying an Issue or a Problem—Sense of Puzzlement)

- Could you tell me how you used GenAI ChatGPT to explore a daunting question or an issue that you found challenging?
- How did GenAI ChatGPT stimulate you to ask questions?
- What issues or questions did you discuss with GenAI ChatGPT that you found relevant and challenging?

Exploration (Analyzing/Asking Follow-Up Questions)

- Can you describe how GenAI ChatGPT helped you examine the problem or issue at hand?
- Did GenAI ChatGPT's responses stimulate your follow-up questions?
- How did GenAI ChatGPT help you brainstorm or search for relevant information?

Integration (Evaluating and Deciding)

- In what ways did GenAI ChatGPT help you integrate new information and ideas with your existing understanding of the topic?
- Can you provide an example of how GenAI ChatGPT helped you better understand the topic or an assignment?
- How did GenAI ChatGPT assist you in synthesizing information and making connections between different ideas or perspectives?

Resolution (Breaking Down Complexity and Contextualized Application)

- Can you tell me about a time when you used GenAI ChatGPT to help you figure out a real-life problem or an assignment?
- How did you use critical thinking to determine if the information provided was helpful or not?
- Have you ever used GenAI ChatGPT to bring together different ideas or perspectives to find an answer to a question to apply it in a real-life situation or work?

Using the PI Critical Thinking Framework

- Does using GenAI ChatGPT require a critical thinker? Please explain your reasoning.
- Thinking back on your experience, how did applying the critical thinking framework (triggering events, exploration, integration, and resolution) I provided you influence your use of ChatGPT?
- Would you say it changed your perspective on ChatGPT from what you believed before?

Appendix B.2. Interview Coding and Thematic Analysis

Table A3. Interview coding and thematic analysis.

Themes	Robe Codes	Emily Codes
1. Guided use of GenAI ChatGPT was associated with reported improvements in critical thinking, as indicated in the four stages of CP. General Codes: 1. Triggering Events Definition: A sense of puzzlement that indicates a curiosity to ask a question or seek a solution for a problem. Then they start to ask GenAI a question to find an answer to the question or better solve a problem.	• Simulating questioning • Piquing curiosity • Motivating interest	• Simulating questioning • Promoting curiosity • Used to focus on unfamiliar areas

Table A3. *Cont.*

Themes	Robe Codes	Emily Codes
2. Exploration: Information on Exchange Definition: Diving and digging deeper to search for more information, look at different ideas, and try to understand the problem better through asking follow-up questions, brainstorming, or asking for some examples.	• Promoting deep thinking • Avoiding matters of broad questions • Asking specific questions matters	• Promoting intrinsic motivation inquiry • Asking for examples, request to provide more details • Deeper understanding of motivation
3. Integration: Connecting Ideas Definition: Trying to make meaning by comparing the responses from GenAI with their background knowledge, social schema, and other external resources for validation, to decide what makes sense and what does not, and whether to accept it or adapt it by adding what they need to reach a final outcome from their collaborative discussion with GenAI.	• Finding answers extremely relevant to the initial question (valuable information) • Gradually narrowing down ideas to match background knowledge • Emphasizing connection of answers	• Providing matching answers to prior knowledge • Background knowledge utilization • Linking new answering to personal background incorporation
4. Resolution: Apply New Ideas Students break down the complexity of the issue, problem, or idea to reach a conclusion or solution. They take everything they have learned and use it to organize their ideas, solve a problem, contextualize it, create something that fits the real situation, and apply it.	• Working outside education, helping dyslexic students in a testing facility • ChatGPT providing a perspective not seen before • Requiring introspection and self-questioning • Promoting metacognitive awareness and idea organization	• Rewriting AI output into personal voice • Using GenAI to structure assignment components • Building on AI responses to develop ideas further • Editing robotic language for clarity and natural tone
2. Positive Impact of the Used Framework on GenAI Use Full cycle of CT: Getting a complete image of how the four stages above impact their use of GenAI.	• Reframing ChatGPT from quick fix to deeper thinking tool • Recognizing AI as a collaborative thinking partner • Emphasizing the need for human Critical thinking with AI	• Shift in perception of GenAI from tool to thinking partner • AI encourages higher-level thinking and better questioning • Recognizing AI's role in supporting educator identity and practice • Emphasis on critical thinking as key to effective GenAI use
3. Critical Thinking Is Indispensable in Using GenAI Definition: This theme emphasizes that while GenAI can support learning, students must bring their own critical thinking to the table. Effective use of ChatGPT requires reflective judgment, evaluation, and the ability to analyze and apply content independently.	• Critical thinking needed	• Reflective evaluative skills needed • Critical thinking needed • Develops thinking process • Evokes critical thinking
4. ChatGPT as a Scaffolding More Knowledgeable Other Educational Tool Definition: Students recognize ChatGPT as a practical educational support tool, especially for writing, note-taking, idea synthesis, and accessing resources. While they note its efficiency and usefulness, they also acknowledge the need for critical evaluation and effective time management when using it.	• Great note-taking tool and synthesizes thoughts • Required more evaluation than expected to find specific valuable information	• User error acknowledgment • Effective time allocation • Efficient classroom resource • Providing valuable resources • Teaching tool in literacy • Potential for interaction • Classroom resource, not teaching tool • Efficiency and timesaving

Table A3. *Cont.*

Themes	Robe Codes	Emily Codes
5. The Nature of the Human–ChatGPT Partnership Definition: This theme captures how students describe their evolving relationship with ChatGPT, not just as a static tool but also as a responsive, personalized platform that adapts to their learning needs and feels more like a collaborative partner than a machine.	• Collaborative scaffolding tool • Personalized learning tool	• Conversation tool • Collaborative interaction tool • Personalized platform

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